

Project Overview

Project: Mission Chief / Leitstellenspiel

Role: Scenario and Systems Designer (2020-2024)

Released: 2013 – present

Format: Web-based Dispatch Center Simulation

Summary: Mission Chief is a dispatch-center simulation built around real-world maps and street routing. Players build their own emergency services with over a hundred unit types and respond to more than 900 operations.

The Care System / SEG-V

Overview

The Care System is based on the German SEG-V (Schnelleinsatzgruppe Verpflegung) and SEG-B (Schnelleinsatzgruppe Betreuung). It simulates affected people and emergency responders at the scene and requires players to dispatch sufficient trained personnel and equipment to support them.

The second part of the system introduced the fire service's *Facheinheit Verpflegung*.

This was one of the final systems I designed from concept through production while working primarily on Mission Chief.

I later advised on adaptations for other regions.

Care Units Responsibility

Affected People

Emergency Responders

Design Challenge

All units vary significantly across Germany and even more internationally. Most are volunteer-operated, and many departments use their own structures and equipment.

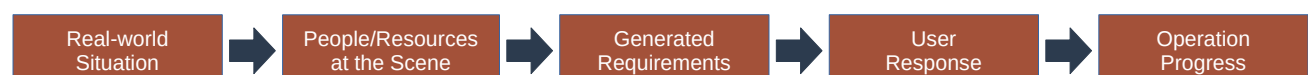
The design therefore had to represent regional practices closely enough to satisfy Mission Chief's realism-focused community while remaining understandable to players from other regions.

The main technical challenge was that demand could not be defined statically per operation: it depended both on the number of affected people and on the emergency responders dispatched by the player. Requirements therefore had to scale dynamically with the situation at the scene.

The system also had to integrate with existing mechanics for vehicles, trained personnel and patients without adding unnecessary UI complexity.

Fire service units introduced an additional constraint: unlike SEG units, they support emergency responders rather than affected people, which needed to be reflected accurately in the simulation.

Workflow Overview



Requirements & Real-World Basis

Real-world care units vary widely because they are organized regionally and rely heavily on volunteers. Some are integrated into units such as the SEG-B, while others serve additional roles within organizations such as the Red Cross or fire service.

Organization is often local rather than standardized, with some communities even relying on supply agreements with supermarkets or local businesses.

Our subject-matter experts were not members of care units themselves, but could contribute their experience interacting with them as emergency responders and people affected by emergencies. Their input helped identify relevant operation types—particularly prolonged emergencies and evacuations—as well as personnel, training and equipment requirements.

One important operational characteristic was that care units are essential to completing some operations without needing to arrive first. The simulation therefore had to allow an operation to begin normally and only halt its progress once care support became necessary.

I supplemented this input by comparing public records and specifications from different emergency services to identify common patterns across regional implementations.

Design Goal

The system needed to be authentic enough that players familiar with these units would immediately recognize their function, while remaining intuitive to existing Mission Chief players.

It also had to stay within reasonable development scope and support later adaptation to other regions.

Solution

Mission Chief operations already present players with requirements that must be fulfilled before an operation can be completed. The new system therefore converts the current situation at the scene into dynamic personnel, vehicle and equipment requirements.

During operation generation, it determines whether affected people will be present and whether emergency responders may require support.

During the operation, it continuously evaluates the number of affected people and simulated responders at the scene. These values generate the current care requirements presented to the player.

I also chose to display both affected people and emergency responder counts, making the relationship between the situation and its requirements visible. Interested players could inspect the conversion from operation values to requirements in the mission details.

Players fulfill these requirements using a crew carrier, equipment truck, kitchen trailer or combined mobile kitchen vehicle.

The truck-and-trailer combination and the dedicated kitchen vehicle represent two common real-world configurations. Supporting both allows players to model different regional setups without requiring separate mechanics.



Formula

For each X affected people and Y emergency responders, the system calculates requirements for A kitchens, B trained cooks and C trained care workers.

For realism, trained personnel only count toward care requirements when arriving with appropriate care units. This prevents personnel qualified for multiple roles from fulfilling two mutually exclusive duties at the same time.



Balancing

Parking space is limited in Mission Chief, so the two kitchen configurations needed comparable gameplay value.

The equipment truck carries additional personnel to compensate for the truck-and-trailer combination requiring two parking spaces instead of one. This also remained plausible within the real-world abstraction.

Special Case: Fire Department

Fire service care units required separate vehicles and training and only support emergency responders. The existing logic was therefore extended with a restricted configuration that ignores affected people.

These units must be evaluated first so personnel restricted to responder support are assigned before SEG personnel are allocated to affected people.

An additional equipment type was also added to support the existing logistics vehicle system (Abrollbehälter).

Trade-offs

A wide variety of real-world vehicles was condensed into a handful of representative options that players could further adapt through existing customization.

Regional training structures were reduced to two qualification levels plus basic training to avoid unnecessary management complexity.

Supply logistics were also abstracted: players manage complete equipment sets rather than individual supplies, consistent with the scope of the wider simulation.

Outcome

The system launched on the German version in two stages: first for SEG units and shortly afterwards for the fire service. Both releases received positive user feedback.

The underlying system was later adapted to other regional versions with only minor additions.